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Genome-wide analysis of SINA family in plants and their phylogenetic relationships[†]

MENG WANG^{1‡}, YING JIN^{1¶}, JUNJIE FU^{1§}, YUN ZHU^{1||}, JUN ZHENG^{2#}, JIAN HU^{1**}, & GUOYING WANG^{1,2}

¹State Key Laboratory of Agrobiotechnology, China Agricultural University, Beijing 100094, P. R. China, and ²Institute of Crop Sciences, Chinese Academy of Agricultural Sciences, Beijing 100081, China

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Abstract

SINA genes in plants are part of a multigene family with 5 members in *Arabidopsis thaliana*, 10 members in *Populus trichocarpa*, 6 members in *Oryza sativa*, at least 6 members in *Zea mays* and at least 1 member in *Physcomitrella patens*. Six members in maize were confirmed by RT-PCR. All SINAs have one RING domain and one SINA domain. These two domains are highly conserved in plants. According to the motif organization and phylogenetic tree, SINA family members were divided into 2 groups. In addition, through semi-quantitative RT-PCR analysis of maize members and Digital Northern analysis of *Arabidopsis* and rice members, we found that the tissue expression patterns are more diverse in monocot than in *Arabidopsis*.

Keywords: SINA genes, evolution, expression, semi-quantitative RT-PCR

Introduction

In plants, proteins containing the SINA domain constitute SINA family. Besides the SINA domain, most SINA family members possess the RING domain (Polekhina et al. 2002). The RING domain locates at N-terminus, whereas the SINA domain locates at C-terminus (Xie et al. 2002). In *Arabidopsis thaliana*, RING-type proteins containing the RING domain were predicted to perform functions as E3 ligases (Stone et al. 2005). The E3 ligase is an important enzyme in ubiquitin pathway. It is responsible for recruiting the target proteins for ubiquitination and considered as the primary substrate recognition component of ubiquitin pathway.

SINAT5, one SINA family member from *Arabidopsis*, has been reported to function as an E3 ligase (Xie et al. 2002). It can interact with NAC1 and make NAC1 modified by ubiquitin. Then, this modified NAC1 is degraded by a 26S proteasome complex (Xie et al. 2002). The degradation of NAC1 results in the decrease of lateral roots. In animal, Siah protein which contains the SINA domain, may function alone as a single subunit E3 ligase, or as a component of the SCF-type complex to facilitate the degradation of many target proteins (Hu and Fearon 1999; Fiucci et al. 2004; Santelli et al. 2005).

In our study, SINA genes were identified from *A. thaliana*, *Populus trichocarpa*, *Oryza sativa* and *Zea mays*. These SINA family members are highly

Correspondence: G. Wang, State Key Laboratory of Agrobiotechnology, China Agricultural University, Beijing 100094, P. R. China. Fax: 86 10 62732012. E-mail: gywang@cau.edu.cn

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[‡]E-mail: wangmeng@cau.edu.cn

[¶]E-mail: jinying518@gmail.com

[§]E-mail: jjfu@cau.edu.cn

^{||}E-mail: xiaozhuhome@163.com

[#]E-mail: zhengjun@cau.edu.cn

^{**}E-mail: hujian@cau.edu.cn

conserved in gene structures and protein motif organization. We used phylogenetic and computational methods to divided SINA genes into two groups. Combining with the expression profiles, we found that each SINA gene may prefer to perform functions in specific tissues. This analysis will be useful in exploring the functions of each SINA gene in plants.

Materials and methods

Plant materials

Maize (inbred line 178) was used for cDNA cloning sequences and RT-PCR analysis. Leaves and roots were harvested from three-leaf-stage seedlings. Mature tassels and immature ears were obtained from mature plants. Total RNA was extracted using a hot phenol method (Kay et al. 1987) and was treated with RQ1 RNase free DNase I (Promega). Approximately 5 µg of total RNA were reverse transcribed using MMV reverse transcriptase (Promega).

Identification of SINA genes in plants

Full-length peptide sequences of SINAT5 (Genbank Accession No. *AAM11573*) was used as a query to do BlastP in the *Arabidopsis* Information Resource (TAIR) (<http://www.arabidopsis.org>). Then all AtSINA protein sequences were used to search the poplar genome assembly version1.1 (http://genome.jgi-psf.org/Poptr1_1/Poptr1_1.home.html) with BlastP. About 10 gene models were identified. Three of them, PtSINA3, PtSINA8 and PtSINA9, haven't full length amino acid sequences. We reannotated genomic sequences of them on the basis of the protein similarity and EST information.

In rice, we used the BLAST programs (BlastP) available on the TIGR Rice database (TIGR rice pseudomolecules release 3) (<http://rice.tigr.org>). The amino acid sequences of all AtSINAs were used as queries. All OsSINA protein sequences were downloaded from TIGR. To identify SINA genes in maize, tBlastN in PlantGDB (<http://www.plantgdb.org>) were performed using all OsSINA protein sequences as queries. According to the information of maize sequences (EST, PUT and GSS), we designed specific primers to amplify maize cDNA. All primers are shown in Table SI. In addition, we have surveyed the sequences encoding the SINA domain in complete or nearly completed sequences of genomes of the moss *Physcomitrella patens* (<http://moss.nibb.ac.jp>, <http://www.ncbi.nlm.nih.gov/blast/mmtrace.shtml>), and the green alga *Chlamydomonas reinhardtii* (<http://genome.jgi-psf.org/Chlre3/Chlre3.home.html>). BlastP and tBlastN were performed using SINAT5 as the query.

Phylogenetic analysis

Multiple sequence alignment was performed using the ClustalX1.83 (Thompson et al. 1997) with default

parameters and was manually adjusted with Jalview (2.07) (Clamp et al. 2004). Phylogenetic tree was constructed using NJ method in MEGA (3.1) (Kumar et al. 2004). The reliability of different phylogenetic groupings was evaluated by using the bootstrap test (1000 bootstrap replications) available in MEGA (3.1).

Motif identification

We identified the protein motifs of 27 SINA genes using MEME (<http://meme.sdsc.edu/meme/meme.html>) with motif length set as 6–210, and e -value $< 1 \times 10^{-10}$. Then the results were manually adjusted. Furthermore, we characterized the motifs using the Conserved Domain Search Service (CD-Search) (Marchler-Bauer and Bryant 2004).

In silico tissue expression analysis in Arabidopsis and rice

The tissue expression profiles of *AtSINAs* were extracted from Genevestigator (Zimmermann et al. 2004) using the AtGenExpression plant tissue. The expression analysis of rice gene was performed using MPSS (20 bp signatures) (<http://mpss.udel.edu/rice>).

Semi-quantitative RT-PCR

In maize, the tissue expression profiles of SINA genes were examined with semi-quantitative RT-PCR. Four tissues, young leaves, young roots, mature tassels and immature ears, were used. Three PCR replications were conducted with the following thermal cycling parameters: 94°C for 30 s, 60°C for 30 s and 72°C for 1 min. Thirty-two numbers of PCR cycles were used in all amplified reactions. The tubulin gene was amplified as the internal control. Table SII shows all primers used in semi-quantitative RT-PCR.

Results

SINA genes in plants

In order to investigate the SINA family in plants, we first collected SINA genes from two eudicots, *A. thaliana* and *P. trichocarpa*, and two monocots, *O. sativa* and *Z. mays*. Multiple searches were used to identify 5 SINA genes in *Arabidopsis*, 10 SINA genes in poplar, 6 SINA genes in rice and 6 SINA genes in maize. All maize SINA genes were confirmed by RT-PCR. These SINA genes were named according to the location in chromosomes and the discoverable orders (Table I).

Genomic organization of SINA genes in Arabidopsis, poplar, rice and maize

For exploring the genomic organization of SINA genes, we first analyzed their exon–intron structures

Table I. ZnF-AN1 genes in *Arabidopsis*, poplar, rice and maize

Species	Gene name	Group	Accession no.*	Locus [†]	Length [‡]
Arabidopsis	<i>AtSINA1</i>	II	NP_181729.1	At2g41980	305
	<i>AtSINA2</i>	II	NP_191363.1	At3g58040	308
	<i>AtSINA3</i>	I	NP_567118.1	At3g61790	326
	<i>AtSINA4</i>	I	NP_194517.1	At4g27880	327
	<i>AtSINA5</i>	I	AAM11573	At5g53360	309
Poplar	<i>PtSINA1</i>	II	fgenes4_pg.C_LG_I000093	LG_I:670238–671581	286
	<i>PtSINA2</i>	II	fgenes4_pm.C_LG_I000095	LG_I:2115588–2117000	296
	<i>PtSINA3</i>	I	gw1.II.2951.1	LG_II:13177725–13180474	330
	<i>PtSINA4</i>	II	estExt_Genewise1_v1.C_LG_III0871	LG_III:17277104–17281238	305
	<i>PtSINA5</i>	II	eugene3.00031885	LG_III:18885651–18887185	311
	<i>PtSINA6</i>	II	estExt_Genewise1_v1.C_LG_VI0616	LG_VI:10945383–10948295	307
	<i>PtSINA7</i>	I	estExt_fgenes4_pg.C_LG_XII0205	LG_XII:1967106–1970129	332
	<i>PtSINA8</i>	I	gw1.XIV.1720.1	LG_XIV:3531188–3533571	330
	<i>PtSINA9</i>	I	gw1.XV.382.1	LG_XV:877405–880050	332
	<i>PtSINA10</i>	II	grail3.0004018901	LG_XVI:3971458–3975863	304
Rice	<i>OsSINA1</i>	I	NP_001042518.1	Os01g13370	347
	<i>OsSINA2</i>	II	NP_001045773.1	Os02g03620	308
	<i>OsSINA3</i>	I	NP_001046599.1	Os02g19140	349
	<i>OsSINA4</i>	II	ABF96019	Os03g24040	301
	<i>OsSINA5</i>	I	NP_001054994.1	Os05g14860	361
	<i>OsSINA6</i>	II	NP_001060528.1	Os07g46560	302
Maize	<i>ZmSINA1</i>	I	EF434384	ZmGSStuc11-12-04.4741.1	300
	<i>ZmSINA2</i>	II	EF434387	ZmGSStuc11-12-04.13685.1	313
				ZmGSStuc11-12-04.11936.1	
	<i>ZmSINA3</i>	I	EF434383	ZmGSStuc11-12-04.12279.1	341
	<i>ZmSINA4</i>	II	EF434385	ZmGSStuc11-12-04.11822.1	302
	<i>ZmSINA5</i>	I	EF434382	ZmGSStuc11-12-04.2391.1	349
	<i>ZmSINA6</i>	II	EF434386	ZmGSStuc11-12-04.26073.1	302

* Accession no. means the protein accession number in *Arabidopsis*, poplar and rice, while it means cDNA accession number in maize;

[†] In *Arabidopsis*, locus represents the AGI locus. In rice, locus represents the TIGR locus. In poplar, locus shows the genomic regions containing the SINA gene. In maize, locus represents the Genome Survey Sequences (GSSs) contigs from PlantGDB; [‡] It shows length of the protein in amino acids.

of coding sequences in *Arabidopsis*, poplar, rice and maize. The exon–intron structures of SINA genes were determined from the genomic sequences of *O. sativa ssp. japonica*, *A. thaliana* and *P. trichocarpa*. The exon–intron structures of the maize SINA genes were determined from PUT and GSS in PlantGDB. As shown in Figure 1, all SINA genes possess three exons except *ZmSINA1*. Moreover, the second exons of most SINA genes show quite strong length conservation (387 bp). For further exploring the exon–intron structures of SINA genes, we examined their intron phases (i.e. the position of an intron within a codon; Nicole et al. 2006). For SINA genes in plants, all introns are in phase 0 (Figure 1).

Besides the exon–intron structures of SINA genes, we also examined their chromosomal locations in *Arabidopsis*, poplar and rice. *Arabidopsis* SINA genes were distributed on four chromosomes, 2, 3, 4 and 5. Two genes were present on chromosomes 3, while chromosome 2, 4 and 5 had only one gene, respectively. Four of the five *Arabidopsis* genes were identified in duplicated blocks (Figure 2A). Members of rice SINA gene family were found to be present on chromosomes 1, 2, 3, 5 and 7 of the rice genome

(Figure 2B). Regions of the rice genome with segmental duplications available at TIGR (http://www.tigr.org/tdb/e2k1/osa1/segmental_dup/index.shtml) were searched in a 100 kb window for the presence of SINA gene family members. One pair was identified in the duplicated segments (Figure 2B). According to the poplar genome annotation in JGI, we found that poplar SINA genes are present on chromosomes 1, 2, 3, 6, 12, 14, 15 and 16 (Figure 2C). Many segmental duplication events have been reported (Tuskan et al. 2006). Based on this result, 5 segmental duplication pairs were identified in poplar SINA genes (Figure 2C).

Conserved motifs of SINA proteins in plants

The conserved motifs of plant SINA proteins were searched using MEME (Bailey and Elkan 1994). A total of eight conserved motifs were identified in all 27 SINA protein sequences (Table II). Through Conserved Domain Search service (CD-Search) (Marchler-Bauer and Bryant 2004), we further characterized these motifs. The motif 1 was identified as the SINA domain, and the motif 2 was identified

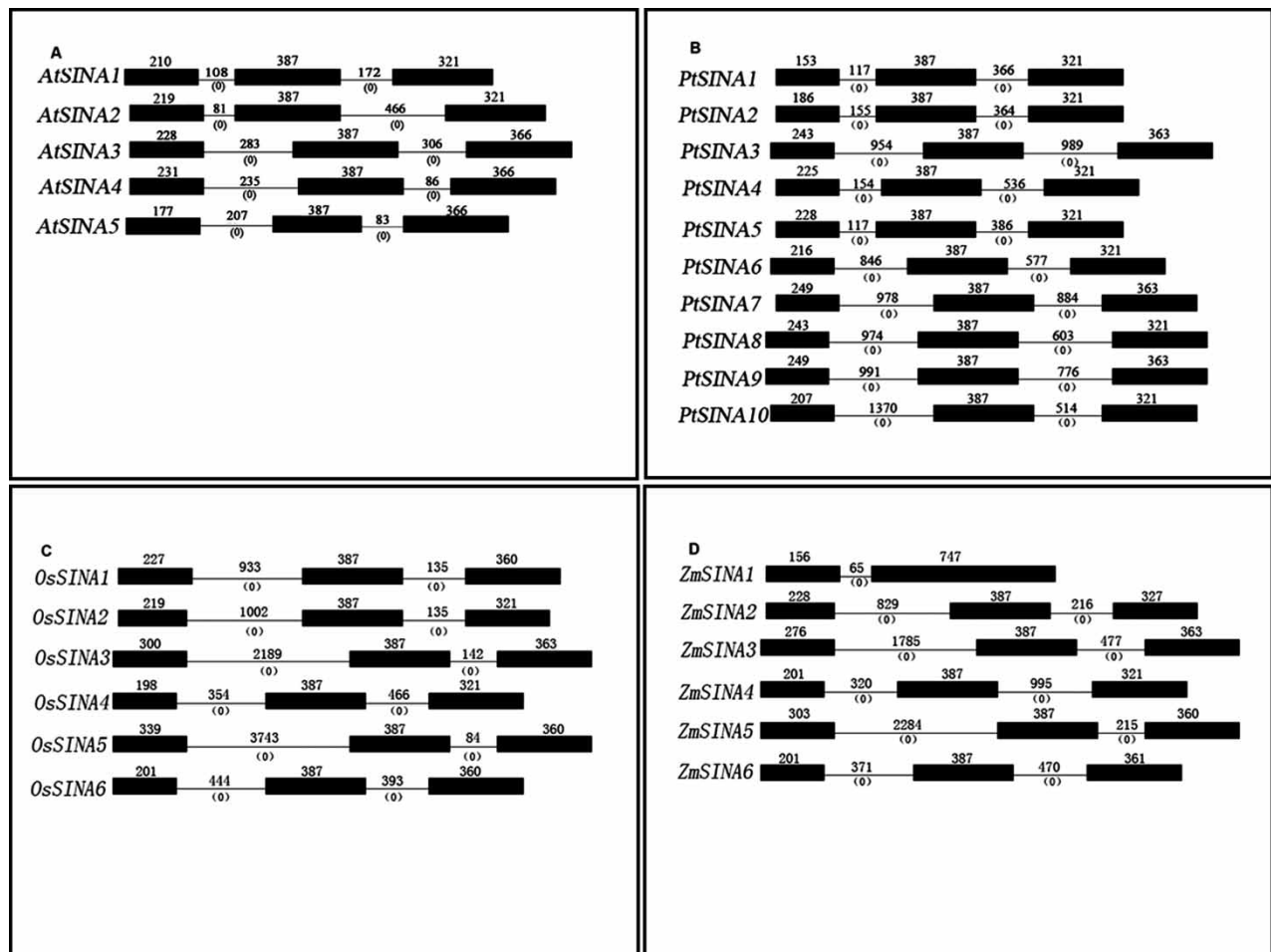


Figure 1. Intron and exon organization of SINA genes. Introns and exons are represented by black lines and black boxes respectively. The length in base pairs of each intron and exon is also indicated. Numbers between brackets correspond to the intron phase. A, B, C and D indicated the intron and exon organization of SINA genes from *Arabidopsis*, poplar, rice and maize, respectively.

as the RING domain. All motifs are shared by monocots and eudicots.

As shown in Figure 3, the N terminal regions are diverse in SINA proteins. Several conserved motifs were identified at N-terminus. At C-terminus, some members have motif 3, while the others do not have this motif. Motif 1 (SINA domain) and motif 2 (RING domain) are existed in all members.

Phylogenetic analysis of SINA genes

For clarifying the phylogenetic relationships of SINA genes in plants, multiple alignment analysis was performed using amino acid sequences of the SINA domain (Figure 4A). The alignment indicated that SINA domains are highly similar. There are many conserved Cys and His residues in the SINA domain, which form 2 zinc finger motifs (Xie et al. 2002).

Based on the alignment, a phylogenetic tree was constructed (Figure 4B), in which all 27 SINA

genes can obviously be classified into two groups. At C terminus, most group I members have motif 3, while all group II members lack this motif (Figure 3). In group II, *PtSINA2* and *PtSINA4*, a duplication pair, have different evolutionary rate. The evolutionary rate of *PtSINA2* gene is higher than the rate of *PtSINA4* gene. This indicated that *PtSINA2* may have gained new functions or lost functions.

Tissue expression analysis of SINA genes in *Arabidopsis*, rice and maize

Through above analysis, it is found that SINA genes are highly conserved. In each species, there are several highly similar SINA genes. These SINA genes may perform functions in different manners. In order to investigate the functional divergence of SINA genes, we analyzed the expression of *AtSINAs*, *OsSINAs* and *ZmSINAs*, in various tissues.

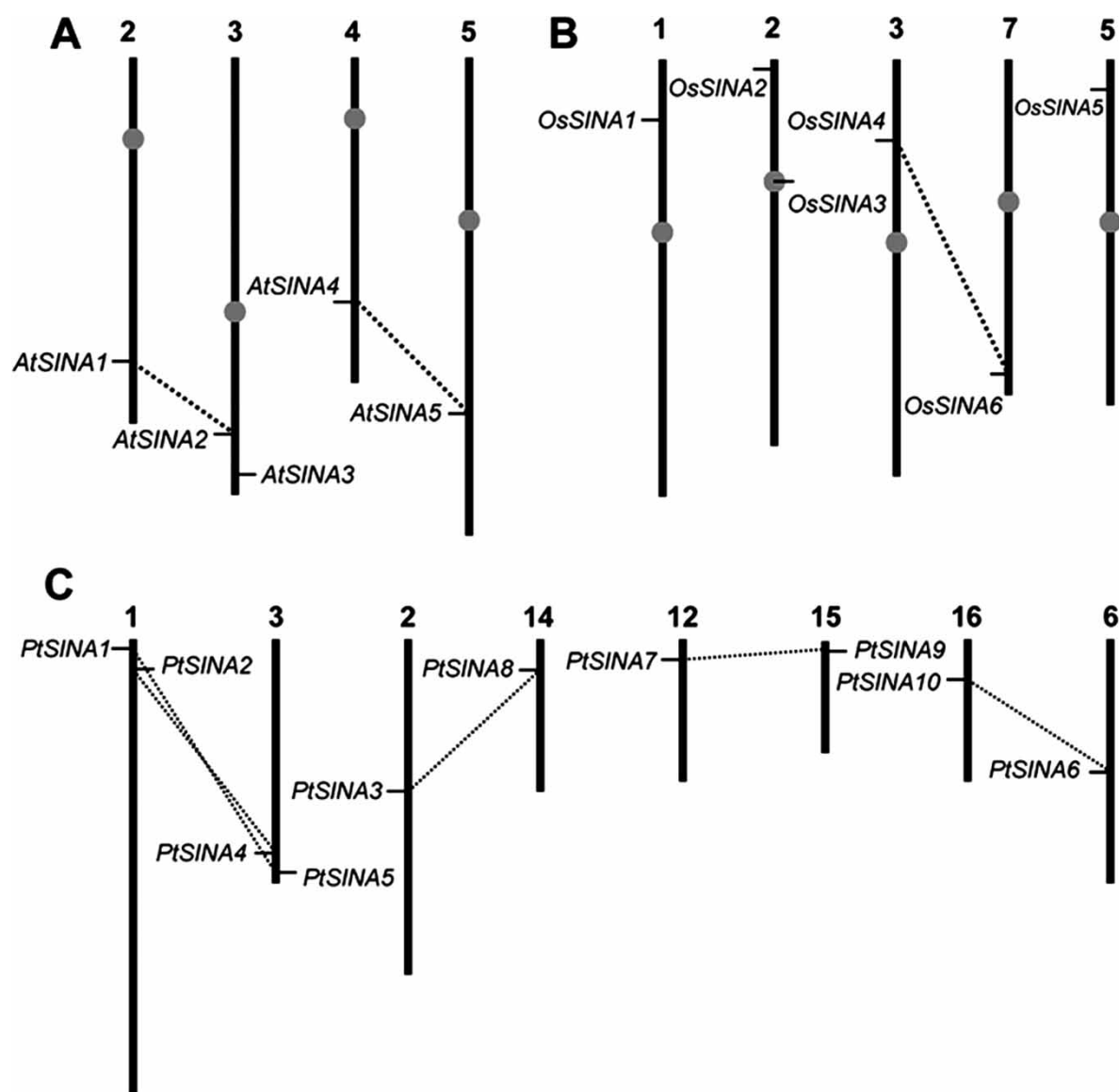


Figure 2. Chromosomal location of SINA genes. The A, B and C show the chromosomal location of SINA family members from *Arabidopsis*, poplar and rice, respectively. Genes present in duplication segments are indicated using dotted lines.

In *Arabidopsis*, the tissue-expression database on Genevestigator was used (Zimmermann et al. 2005). The data showed that *AtSINA* genes were expressed in the most tissues examined (Table III). All *AtSINA*s are

highly expressed in pollen, except *AtSINA3*. The tissue expression patterns of *AtSINA1* and *AtSINA2*, a duplication pair, are different, which suggested the possibility of sub-functionalization of this gene pair.

Table II. Conserved motifs identified by MEME using amino acid sequences of 27 SINA genes.

Motifs	Multilevel consensus sequence
1	RQELGNIRCLALEKVAESLELPCKYQSLGCPEIFPYYSKLKHEQQCNFRPYNCPYAGSECSVAGDIPYLV AHLK-DDHK VDMHDGCTFNHRYVKS NPREVENATWMLTVFHC FGQYFCLHFEAFQLGMAPVYMAFLRFMGDE-NEARNYSYSLE VGGNGRKMTWQGTPRSIRDSHRKV RDSHDGLIQRNMALFFSGGDRKELKLRVTGRIWK
2	CPVCTNSMYPIHQCPNGHTLCSTCKTRVHNRCPTC
3	DDGACIPNLCS
4	DMDSIECVSPDST
5	DEDEIHNNHQQH
6	QPSVSKPHNTNNNTNNNNN
7	ASVTYIDDSHSEVIDPPKTE
8	APGGSIFKEVLESDSTGDD

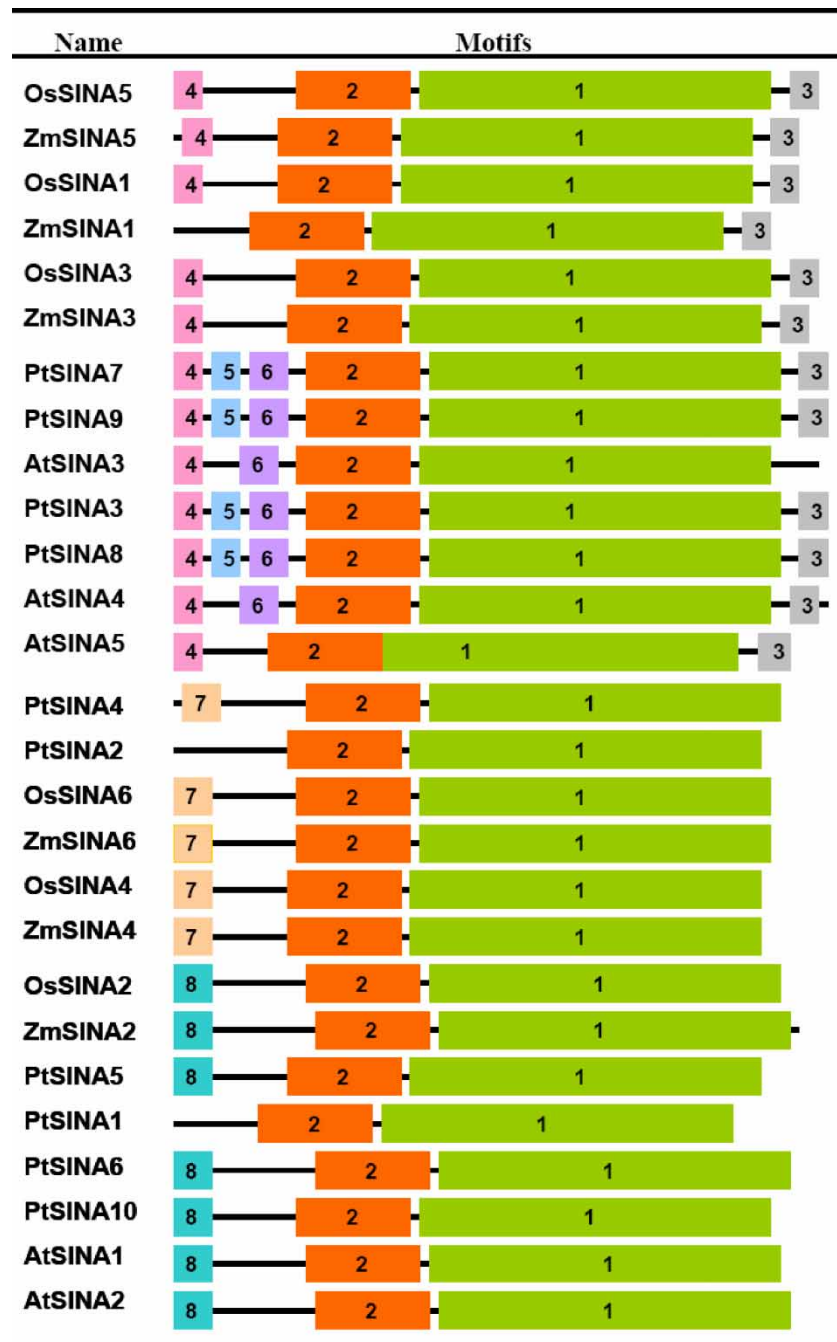


Figure 3. Motif organization of ZnF-AN1 proteins in plants. Motifs were identified by MEME using amino acid sequences.

In rice, the tissue-specific expression profiles of *OsSINA* genes were analyzed using the publicly available data from rice MPSS (massively parallel signature sequencing) (<http://mpss.udel.edu/rice/>). The data showed that *OsSINA1* and *OsSINA5* expressed in all tissues at various levels, while *OsSINA4*, *OsSINA2* and *OsSINA3* expression is excluded from several tissues (Table IV). Meanwhile, *OsSINA6* expression cannot be detected in any tissues examined. In addition, the pollen expression patterns are different in *Arabidopsis* and rice. In rice, only one gene, *OsSINA1*, is highly expressed in pollen.

Other rice SINA genes, *OsSINA2*, *OsSINA3*, *OsSINA4* and *OsSINA5* are highly expressed in mature stigma, immature panicle, leaf and stem, respectively. For the duplication pair, *OsSINA6* and *OsSINA4*, their tissue expression patterns are obviously different (Table IV).

In maize, we used semi-quantitative RT-PCR to examine the transcript levels of *ZmSINAs* in four tissues, roots, leaves, ears and tassels. As shown in Figure 5, the tissue expression patterns of *ZmSINAs* are diverse. Three members, *ZmSINA1*, *ZmSINA5* and *ZmSINA6* are highly expressed in tassels, one

[illegible]

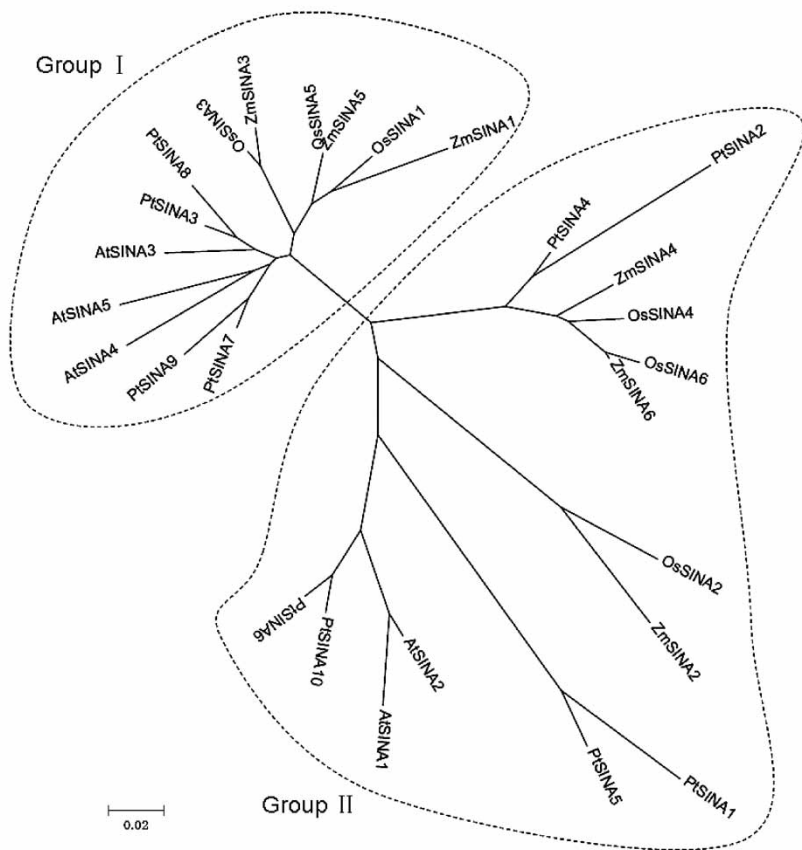


Figure 4. A, The multiple alignment of amino acid sequences of the SINA domain using ClustalX1.83. The locations of two zinc fingers are indicated above the alignment. The conserved residues of Cys and His are showed above the alignment by the * symbols. B, An unrooted phylogenetic tree. The phylogenetic tree was constructed using the NJ method and a bootstrap test with 1000 replicates.

member, *ZmSINA2*, is highly expressed in ears, and two member, *ZmSINA4* and *ZmSINA5*, are highly expressed in leaves. The expression level of *ZmSINA3* is very low in all the tissues examined.

Discussion

The evolutionary process of SINA genes in plants

In order to investigate the history of SINA genes in plants, we surveyed the sequences encoding the SINA domain in the green alga (*C. reinhardtii*), and the moss (*P. patens*). We identified one non-redundant sequence (Genbank Accession No. *Bf972724*) encoding

the SINA domain in the moss, but we found no SINA domain sequence in the green alga. In addition, the SINA protein of moss does not possess the motif 3, which suggested that the occurrence of group I members was possibly posterior to the generation of group II members.

Relative to the moss, the expansion rate of SINA genes in angiosperm is higher. Our study revealed that SINA genes in plants are multigene family with 5 members in *Arabidopsis*, 10 members in poplar, 6 members in rice and at least 6 members in maize. In *Arabidopsis*, poplar and rice, none tandem duplication events were found, while several segmental duplication events occurred (Figure 2). Some

Table III. *In silico* tissue expression analysis of SINA genes in *Arabidopsis*.

Gene	Callus	Cell suspension	Seedling	Carpel	Pollen	Seed	Stem	Rosette	Roots
<i>AtSINA1</i>	236	239	427	464	3525	1118	263	507	268
<i>AtSINA2</i>	1410	1178	956	640	4813	1301	1067	853	1185
<i>AtSINA4</i>	1413	1062	1117	905	43,076	1692	1532	1106	1190
<i>AtSINA5</i>	1692	1033	691	512	32,393	1544	1003	655	813
<i>AtSINA3</i>	2212	1923	1509	857	1054	1518	1588	1241	1496

Microarray data from the AtGenExpress organ dataset was extracted using Genevestigator. The relative expression level is indicated by absolute signal values.

Table IV. *In silico* tissue expression analysis of SINA genes in rice.

Gene	Germinating seed	Immature panicle	Vegetative meristematic tissue	Stem	Young leaf	Young root	Ovary and mature stigma	Mature pollen
<i>OsSINA1</i>	4	7	53	55	20	16	106	279
<i>OsSINA2</i>	29	0	0	55	2	15	97	0
<i>OsSINA3</i>	8	91	0	3	0	0	1	3
<i>OsSINA4</i>	0	2	16	0	236	0	79	43
<i>OsSINA5</i>	270	457	387	1072	177	72	186	121
<i>OsSINA6</i>	0	0	0	0	0	0	0	0

The transcript abundance was obtained from rice MPSS libraries. The relative expression level is indicated by transcripts per million (TPM).

duplication pairs, for example, *PtSINA6/10* pair, have similar evolutionary rate, suggesting the possibility of sub-functionalization of these gene pairs. Some duplication pairs, for example, *PtSINA2/4* pair, have different evolutionary rates, which suggested *PtSINA2* having rapid evolutionary rate may have gained new function or lost function.

After gene expansion, in angiosperm, high selective pressure made SINA genes extremely conserved. Most SINA genes have three exons in coding sequences. Moreover, the second exon and third exon are more conserved. In the majority of SINA genes, 387 bp nucleic acids constitute the second exon. Not only the gene structures, but also the protein motif organization, are very similar. The RING domain and SINA domain are more conserved. However, in SINA family, the N terminal regions are diverse (Figure 3).

The tissue expression patterns are more diverse in monocot than in Arabidopsis

In *Arabidopsis* (eudicot), most SINA family members (4/5) are highly expressed in pollen (Table III).

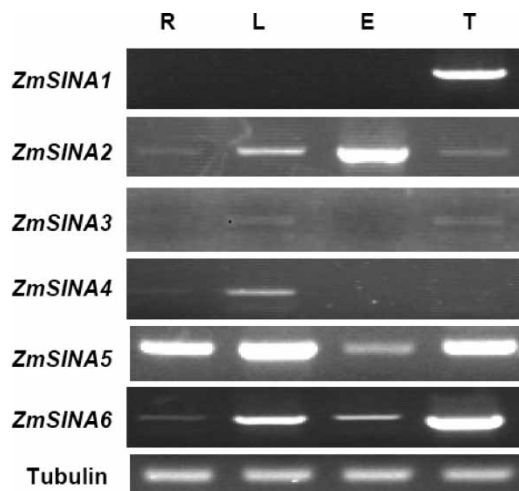


Figure 5. Semi-quantitative RT-PCR analysis of SINA genes in maize. In maize, the tissue expression profiles of SINA genes were examined with semi-quantitative RT-PCR. The tubulin gene was amplified as internal control. R, young roots; L, young leaves; E, immature ears; T, mature tassels.

In maize (monocot), half of members are highly expressed in tassels, and the expression of one member is ear-specific (Figure 5). In rice (monocot), the tissue expression patterns of SINA genes are more diverse. *OsSINA1*, *OsSINA2*, *OsSINA3*, *OsSINA4* and *OsSINA5* are highly expressed in pollen, mature stigma, immature panicle, leaf and stem, respectively. This result indicated that monocot SINA genes probably play more roles or maybe perform the same function in specific tissues. In addition, the tissue expression patterns of *ZmSINAs* are similar to *OsSINAs*. Such as, the expression of *ZmSINA2* and *OsSINA2* are female-flower-specific, and the expression of *ZmSINA4* and *OsSINA4* are highly expressed in leaves.

Putative functions of plant SINA genes?

In *Arabidopsis*, proteins containing RING domain were predicted to take part in ubiquitin pathway as E3 ligases (Stone et al. 2005). SINAT5 (AtSINA5), one RING-type protein, can interact with NAC1 and make NAC1 degraded through ubiquitin pathway (Xie et al. 2002). The degradation of NAC1 makes the lateral root decrease. In this process, SINAT5 acts as an E3 ligase (Xie et al. 2002). In animals, Siah, containing the RING domain and the SINA domain, may function alone as a single subunit E3 ligase to degrade many proteins, such as the netrin-1 receptor/deleted in colorectal cancer, DCC; the nuclear receptor corepressor, N-CoR; the motor protein, Kid; the transcriptional activator, OBF-1; and so on (Hu and Fearon 1999; Fiucci et al. 2004). The N-terminus and RING domain of Siah bind E2 proteins, and C-terminus including SINA domain can be considered as a substrate-interaction region (Polekhina et al. 2002). Moreover, The RING domain has been shown to assist in the *in vitro* transfer of ubiquitin from E2 to substrates (Lorick et al. 1999). In addition, Siah has also been shown to interact in an SCF-type complex including Skp1, Ebi, Siah interacting protein (SIP), and adenomatous polyposis coli protein (pAPC) to facilitate the degradation of β -catenin in a p53-dependent manner (Santelli et al. 2005).

In plants, other SINA members may also act as E3 ligases to perform functions. Because the C-terminus

can interact with substrates, group I and II members probably interact with different substrates. It has been demonstrated that one E3 ligase can interact with many substrates (Smalle and Vierstra 2004). In group I and II, each member may be specialized to be prone to interact with one class of substrates. This possibly made plants more advantageously adapt to more diverse conditions.

Whether some factors can interact with SINA in plants, and how these processes carry through still require further investigation.

In a conclusion, we analyzed SINA families in *Arabidopsis*, poplar, rice, maize and moss. And we identified 5 members in *Arabidopsis*, 10 members in poplar, 6 members in rice, 6 members in maize and 1 member in moss. The gene structures and protein motif organization of SINA genes are highly conserved in plants. According to the motif organization and phylogenetic tree, SINA family members were divided into two groups. After gene expansion, in angiosperm, each SINA genes probably performed functions in different tissues. They may act as E3 ligases to work.

Acknowledgements

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Appendixes

Table SI. The primers used to amplify the cDNA of maize SINA genes.

Gene name	Gene specific primers
<i>ZmSINA1</i>	ATCCCCTCCGTGTCCCGGAGC GATCAGCCGGCCACCCGTAGG
<i>ZmSINA2</i>	AGTGGTGGGCTGCTGATGGT GGATGATGGGATGGAATGGAA
<i>ZmSINA3</i>	AATCTGAGCCTCGGCACAAG CCAACAGTAAATACGCCCTGAC
<i>ZmSINA4</i>	GTGCGTAGCCTGCCCATTTC GGTAGCAACCTCCAACCTCT
<i>ZmSINA5</i>	CCAATCCAGGCGAAACAAAC CAATCGTCTCCCTTGCTCAG
<i>ZmSINA6</i>	TATCGCATCACGCCAAAGCC ACCCACGACTCAAAGATTCCAAC

Table SII. The primers used to perform semi-quantitative RT-PCR of maize SINA genes.

Gene name	Gene specific primers
<i>ZmSINA1</i>	GCTGACGGTGTTCCACTGCT GATCAGCCGGCCACCCGTAGG
<i>ZmSINA2</i>	AGTGGTGGGCTGCTGATGGT GGATGATGGGATGGAATGGAA
<i>ZmSINA3</i>	CTTATGCTGGTTCTGAATGCTC CCAACAGTAAATACGCCCTGAC
<i>ZmSINA4</i>	CGCTATGTAAAGCCAAACCC GGTAGCAACCTCCAACCTCT
<i>ZmSINA5</i>	CTGGCTCTGAATGTGCTGTGG CAATCGTCTCCCTTGCTCAG
<i>ZmSINA6</i>	ACACGGATCACAATGCCAATAC ACCCACGACTCAAAGATTCCAAC
Tubulin	GCTATCCTGTGATCTGCCCTGA CGCCAAACTTAATAACCCAGTA